



Contents lists available at ScienceDirect

Transportation Research Part D

journal homepage: www.elsevier.com/locate/trd

Lack of resilience in transportation networks: Economic implications

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ARTICLE INFO

Keywords:

Risk
Regional economic modeling
Network science
Transportation networks
Resilience
Policy

ABSTRACT

Disruptions to transportation networks are inevitable. When road networks are not resilient, or in other words, do not recover rapidly from disruptions, unpredictable events can cause significant delays that may be disproportionately greater than the extent of the disruption. Enhancing transportation system resilience can help mitigate the consequences of disruptions; however, required investments are difficult to justify given the low probability of such events. This paper calculates economic implications of unmitigated random disruptions in urban road systems. We modeled delays in transportation networks and demonstrated how resilience can be integrated into macroeconomic modeling via the transportation planning model, REMI Transight. The model was applied to 10 cities in the United States to forecast the impact of disruptions on gross domestic product (GDP). Different disruption scenarios were modeled and the magnitude of disruption was used to calculate additional delays in transportation networks, which were then integrated into the Transight model. The results were compared to a baseline case, where economic impact was assumed to be proportional to the magnitude of disruptions. Results show that losses in GDP were far more pronounced in the case scenario as compared to the baseline. The losses tended to be higher in wealthier and more economically productive cities. The economic output tends to rebound one to two years after a disruptive event. We conclude that different topology in transportation networks in different cities requires explicit consideration and quantification of resilience to support investment decisions designed to improve transportation networks in cities.

1. Introduction

Currently, most mandated development-related transportation planning is intended to prepare for frequently occurring and observable disruptions, while unpredictable events that have not yet occurred attract less attention. The current norm for improving transportation networks and remedying the economic impact of delays is undertaken through management of the network for specific threats to reduce the travel time and improve efficiency motivated by the goal of reducing delay (Belenky, 2011). Current practice is to evaluate road performance with Level of Service (LoS), or a similar measure of efficiency, during the worst traffic of an average day and when the whole road network is running as expected (Highway Capacity Manual, 5th Edition, 2010). This emphasis on travel time and the monetary value of its duration allows prospective projects to enter the realm of cost-benefit analysis but carries the

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<https://doi.org/10.1016/j.trd.2020.102419>

shortcoming of representing only observable and predictable events. Recent experiences clearly show that a focus on resilience is necessary. The economic impacts of resilience (or lack thereof) in transportation networks have not been studied, but will ultimately be essential to justify future investment. The National Academy of Sciences defines system resilience as the capacity to be prepared for an event, absorb its shock, and then recover and subsequently adapt following the event (NRC, 2012). Lounis and McAllister (2016) quantify resilience as the change in a system's performative function and quality over the course of either or both an extreme event or a gradual accumulation of internal or external stress. Here, and in previous publications (Ganin et al., 2017, 2019), we consider transportation resilience to be a property of the network that allows for fast recovery of travel to expected speeds, despite disrupted roads and intersections in the network. Cumulative delays associated with disruptions can be considered a metric of network resilience.

The transportation field recognizes that variances in expected travel times have a cost, even when not incurred, as they need to be planned for by travelers. This is known as the Value of Reliability (VOR) (U.S. Department of Transportation, 2016). While VOR is a well-researched field and methodologies to estimate it do exist (Fosgerau and Karlström, 2010; Lam and Small, 2001), no standardized method has yet been adopted in the U.S. (U.S. Department of Transportation, 2016). The cost of unreliable transportation may not be consistent across sectors as some workers have more flexibility in where and when they work than others. These costs may also fluctuate with the amplitude of the delay, with small delays potentially being negligible, though more research is needed (Fosgerau et al., 2007; Mackie et al., 2003; U.S. Department of Transportation, 2016). A more standard metric used in planning is Value of Time (VOT), which is generally calculated on the assumption that variance in travel time from one scenario to another is certain. VOT is normally linearly calculated based on wage rates (U.S. Department of Transportation, 2016). Both of these methods fail to capture the costs of unpredictable delays, which cannot be accounted for in schedules and may have different costs than VOR or VOT.

The realm of possibilities between average traffic conditions and unpredictable disruptions, such as natural disasters, is comprised of events that can have significant disruptions to mobility, and yet are outside the normal realm of planning and analysis. These events may be unpredictable in both space and time and unknown in nature, to the point of being random. Therefore, these unpredictable events should be the focus of resilience inquiries here and elsewhere (Ganin et al., 2019, 2017). While mitigating the risk of disruption (i.e., strengthening key nodes and links in transportation networks) is appropriate for specific hazards, events that are highly uncertain in space and time challenge our ability to characterize vulnerability to them and implement effective risk mitigation measures. Cost concerns ensure that completely minimizing physical risk at one location may inherently limit our ability to reduce risk elsewhere. Similarly, hardening a transportation system against the risk of new types of disruptions, such as cyber-attacks on Intelligent Transportation Systems (ITS), is difficult because the potential range of risk is too vast, or poorly understood, to be effectively predicted (Ganin et al., 2019). Network-wide management is therefore more appropriate than location-specific solutions, where the objective is to keep people and goods flowing through the network in spite of disrupted parts of the network. An apt measure of a network's performance with respect to that objective is resilience. The capabilities and characteristics that should be fostered to deliver system resilience in the face of disruptive events are currently studied in many disciplines, including transportation (e.g., Cox et al., 2011).

Across various contexts, there is a growing recognition that lack of resilience can have grave socioeconomic consequences, especially in the context of damaged interconnected infrastructure (Florin and Linkov, 2016). Such is the finding of a recent World Bank report on infrastructure, as a key enabler of economies and the macroeconomic impacts, not being resilient (Hallegatte et al., 2019). Presidential Policy Directive 21 – Critical Infrastructure Security and Resilience (Obama, 2013) formalized the call to enhance the nation's critical infrastructure functioning and resilience by recognizing the importance of operable critical infrastructure, including transportation systems. This resilience-related Policy Directive focused on 16 sectors that were considered vital to national economic security, public health, and safety ("Critical Infrastructure Sectors," 2019). The current state of practice in transportation planning lends itself to the conclusion that advances in resilience research need to be integrated into planning norms to help account for uncertain events and emerging risks. Quantitative resilience modeling results can be used in tandem with efficiency-driven modeling efforts to conduct tradeoffs among multiple objectives in transportation network investment.

The extent of socioeconomic impacts due to disrupted infrastructure is a currently researched field, yet mostly limited to forecasting or assessing impacts of specific disruptive events. For example, Ham et al. (2005) assessed the anticipated economic implications if an earthquake were to occur in the New Madrid Seismic Zone in the U.S. Midwest. They concluded that the ensuing disruption to U.S. commodity flow could pose a significant threat to economic stability and recovery at the regional, national, and international scales. Ham et al. (2005) define transportation network resilience as referring to the adaptability of commodity flow such that goods can be transported via multiple modes. Similarly, Tatano and Tsuchiya (2008) developed a spatial computable general equilibrium (CGE) model to estimate economic losses (e.g., changes in the cost of travel time) attributed to earthquake disruption of freight and passenger transportation flow. They used the 2004 Niigata-Chuetsu Japan earthquake as a case study, and regional economic losses were measured as a function of inter- and intra-regional trade (Tatano and Tsuchiya, 2008). Pelling et al. (2002) discussed how the 1995 Kobe, Japan earthquake increased transportation costs in the region by over 50% and increased the cost of goods in the region by 10%. Internationally, the disablement of the Kobe Port halted the import and export of goods. Pelling et al. (2002) suggest that disasters possess an "inflationary potential" due to their capacity to affect the "production, distribution, marketing, and consumption" functions of markets. Cho et al. (2015) show how disruptions to critical highway infrastructure, such as highway bridge and tunnel damage, can lead to economic losses in the U.S. on a state-by-state and industry basis. They conclude that the states and industries that can adapt to disruption suffer the least economic loss. Therefore, redundancy in transportation networks can mitigate the consequences of a disruptive event. This conclusion is mirrored by Worton (2012), who suggested that resilience engineering should focus less on efficiency than on the capacity for preparedness, recovery, and adaptation (Mattsson and Jenelius, 2015). The shortcomings of existing studies on the economic implications of transportation disruptions are twofold: (1) resilience is

often conflated with potential economic damages and (2) economic modeling is not underpinned by any analysis of transportation network topology.

This paper aims to demonstrate that joining resilience analysis with regional economic modeling can advance the methodological approach necessary for planning. This integration will allow for transportation costs to be associated with travel time delays, as delays are a primary contributor to economic impact. The current paper uses REMI TranSight (TranSight 4.2 User Guide, 2018), an established economic modeling process that uses input–output, computable general equilibrium, econometric, and economic geography methods. Disruption-induced delay results generated by Ganin et al. (2017) were translated to economic outcomes by formulating them as inputs to REMI TranSight. This expanded the ways to use travel delay information, which is more traditionally used in scenario and policy analysis. We introduce a process for quantifying the economic implications of resilience, or lack thereof, and demonstrate this process for 10 U.S. cities. This can be used to develop a planning approach that explicitly considers resilience. The methodology used in this study can be instrumental in the transition from current risk-based planning to true resilience planning, supported by economic analysis and subsequent selection of management alternatives.

2. Methods: Integration of transportation network resilience and economic models

This paper joins two independently developed and documented models to assess the economic impacts of resilience in transportation networks: (1) Ganin et al.'s (2017) urban traffic network simulation, which estimates travel times given either a predictable disruption (e.g., peak commute hours) or random disruptions and accidents (e.g., natural disasters and major accidents), and (2) TranSight (TranSight 4.2 User Guide, 2018), a regional economic forecasting model oriented specifically for simulating the outcomes of changes in transportation systems.

2.1. Network model for simulating delays associated with disruptions

Effects of disruptions on an urban transportation infrastructure are taken from Ganin et al. (2017). Specifically, Ganin et al. (2017) modeled travel delays that result from road network disruptions during peak-hours for private vehicle commuters in 40 urban areas in the continental United States.

In order to study travel delays under both normally functioning and disruption scenarios, Ganin et al. (2017) first built transportation graphs (as in Fig. 1) for each of the urban areas, comprised of intersections connected by roadways, and then simulated commuter trips on the graphs. Specifically, urban areas were spatially divided based on the Voronoi tessellation method¹ and were centered at each road intersection. The expected population of each cell was calculated using data from the U.S. Census Bureau. Trip distribution was accomplished with a modified gravity model and the privately-owned vehicle was the only mode of travel studied. Route assignment was done assuming free-flow speeds on all roadways. Further, commuters were assumed to travel along the routes which minimized their trip durations. The output of the model is the annualized travel time per commuter. Free parameters of the model are calibrated to match measured data on the annualized delay per a peak-hours auto commuter, as given by the Urban Mobility Scorecard (Schrunk et al., 2015).

To characterize the resilience of modeled cities, Ganin et al. (2017) generated disruptive events on the transportation networks by disabling links. Resilience was quantified as the additional delay resulting from these disruptions. More specifically, Ganin et al. (2017) randomly selected a fixed fraction ρ of network links that was made non-functional by reducing the free-flow speeds of the network links to 1 km/h. Links were selected at random, with probabilities proportional to their lengths. Then, traffic was redistributed per the updated link free-flow travel times, assuming the same origin–destination demand matrix and assuming travelers were still using routes that minimized trip duration. Additional delays that resulted from these events were evaluated as the difference between annualized travel time with and without a disruption. The details of each step of the model development can be found in Ganin et al. (2017).

2.2. Economic modeling to forecast the impacts of transportation network changes

TranSight (TranSight 4.2 User Guide, 2018) is designed to be used with transportation forecasting models to translate the outcomes of improvement measures into regional economic implications. Regional Economic Models Inc. (REMI), the developer of TranSight and related products, maintains models for a wide variety of U.S. regions and states in order to support research. For a particular city or region, the effects of transportation projects are forecasted in economic terms that include gross domestic product (GDP), employment, delivered price, commodity access, labor access, and relative cost of production. REMI mixes techniques from Input-Output (I-O) and Computable General Equilibrium (CGE) modeling, as well as economic geography and econometric techniques (Model Equations, 2017). In this research, TranSight is paired with forecasted delays that result from transportation system disruptions instead of changes in travel times that are associated with road network improvements.

The inputs to TranSight from a transportation model include: (a) changes to Vehicle Miles Traveled (VMT), (b) changes to Vehicle Hours Traveled (VHT), and (c) vehicle trips attributable to improvement measures. Changes in velocity (measured by VMT divided by VHT) from the baseline scenario to the improvement (test) scenario are formulated as having a proportional effect on *transportation*

¹ Voronoi tessellation is a technique for partitioning a plane into cells around points such that the distance from a given point to the boundaries of the cell is less than that of the other points (Skiena, 2008).

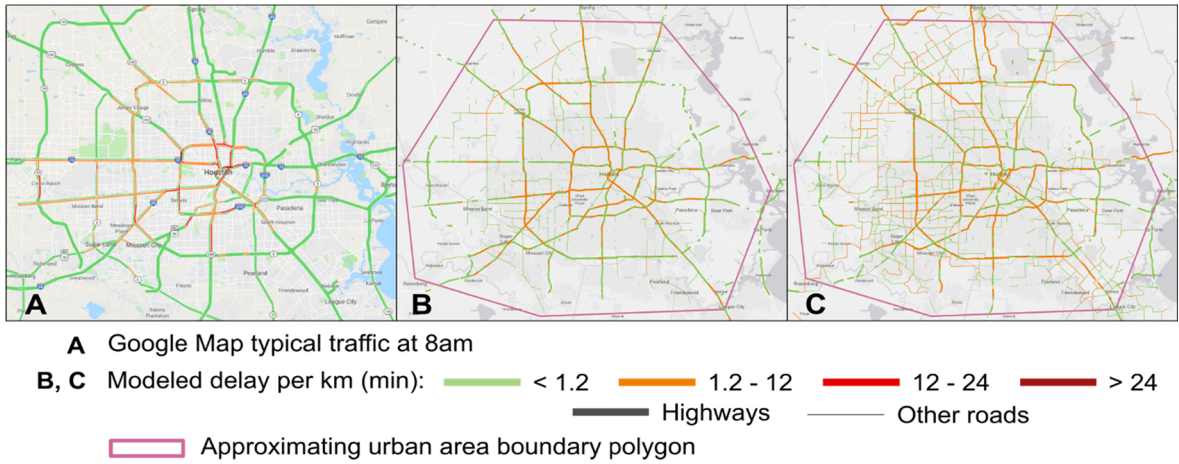


Fig. 1. Roadway networks of Houston, TX: (A) Congestion patterns at 8 am per Google Maps, (B) Modeled delays per kilometer of travel in normal conditions, and (C) Modeled delays per kilometer of travel under 5% disruption. The purple line shows an approximation of the urban area boundary in panels (B) and (C). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

cost and changes in trips that can be made in a given amount of time (measured by the number of trips divided by VHT) are formulated as having a proportional effect on *accessibility cost*. Changes from the baseline transportation network are presumed to affect various economic variables via changes in “effective distance,” which functions to change travel time or commuting time and expenses. Cost-savings due to reduced travel times accrue to industry firms in the model due to reduced commuting and transportation costs and increased access to markets.

Cost-savings, capital investment, and other financial and economic concerns associated with prospective infrastructure projects are related to the regional economy. Changes to economic variables are modeled via policy variables that represent the effect of travel time on individual spending on fuel and subsequently disposable income and consumer spending. Additionally, costs to industries are related to transportation via the extent to which labor demand can be met and the composite price of goods sent to market. All of the linkages between travel times and associated costs to the policy variables that are used in TranSight are detailed in [Model Equations \(2017\)](#) and the [TranSight 4.2 User Guide \(2018\)](#).

2.3. Connecting transportation network resilience and regional economic modeling

For the specified road network disruption severity ρ , quantifying the fraction of affected roadways, the network model ([Ganin et al., 2017](#)) calculates the resulting average annualized travel time $T(\rho)$ per a peak-hours commuter, which can be compared to travel time in the absence of disruption $T(0)$. The topological attributes that yield higher or lower $T(\rho)$ values are not called into question in this research. We assumed a linear relation between the transportation costs and the relative increase in travel time due to network disruption, estimating the corresponding percent increase in transportation costs $c(\rho)$ as

$$c(\rho) = \frac{\Delta T(\rho)}{T(0)} \quad (1)$$

where $\Delta T(\rho) = T(\rho) - T(0)$.

For the purposes of this demonstration, we relied on changes in gross domestic product (GDP) as an indication of economic impact for individual cities assessed. We did not probe the specific mechanisms of the economies of each city. To quantify the effects of the transportation cost increase on GDP, we utilized the TranSight model, and then generated the relative change in GDP as a function of disruption severity for the 10 cities of interest. Two scenarios are modeled:

(1) *Baseline scenario.* Assumes that transportation cost increases are directly proportional to road disruption severity:

$$c^0(\rho) \equiv \rho, \quad (2)$$

(2) *Test scenario.* Assumes that transportation cost increases are proportional to the additional travel time induced by disruption, not to the fraction of roads affected (see Eq. (1)).

3. Results

[Table 1](#) displays the percent increase in transportation cost computed for different cities at specific values of road network disruption severity ρ and serves as the basis for subsequent analysis of economic implications.

We observed that the same disruption results in different cost increases for different cities (note that cost is proportional to additional delay). The cost escalated quickly – even at 3% disruption, San Francisco exhibits a cost increase of 34%. The economic

Table 1

Transportation cost $c(\rho)$ increases by city that result from road network disruptions of severities varying from 1 to 5%. Shown in bold are the transportation cost increase values exceeding 25%.

		Fraction of Affected Roadways (Network Links), ρ				
		1%	2%	3%	4%	5%
Transportation Cost Increase, $c(\rho)$	Atlanta	4%	10%	16%	23%	33%
	Detroit	3%	6%	9%	14%	19%
	Houston	5%	11%	16%	24%	32%
	Jacksonville	7%	13%	22%	33%	44%
	Los Angeles	1%	3%	5%	7%	9%
	Miami	4%	9%	13%	18%	23%
	Orlando	4%	9%	14%	20%	26%
	San Francisco	9%	20%	34%	43%	51%
	Seattle	3%	6%	9%	13%	17%
	Tampa	6%	12%	20%	26%	37%

impacts of transportation disruption were most pronounced in San Francisco, where 5% disruption resulted in a 51% increase in transportation costs. Similar disruption in Los Angeles resulted in only a 9% increase in transportation costs. Jacksonville is overall the second most severely impacted city considered, with a 44% increase in transportation costs.

TranSight forecasts future economic performance in order to understand the short- and long-term implications of disruptions and management policies. Fig. 2 shows the impact of a 5% increase in transportation costs on the GDPs for 2019 and the forecasted subsequent five years for the baseline (2a) and test case (2b) scenarios. All simulated cities showed significant impact on GDP in 2019, the year in which the disruption occurred; the residual effect of that shock over the five subsequent years, however, was relatively small. No remediation or recovery activity was simulated and transportation costs returned to prior levels in 2020–2024. All of the cities recover to within 0.2% of their expected GDP (simulated GDP in the absence of a shock) by 2020 and have exceeded it by 2023. The profile of the initial impact and recovery was very similar for the test case, yet the magnitude was substantially greater, with disruption roughly an order of magnitude worse than the baseline case. Notice the difference in the y-axis of the Fig. 2 graphs.

In Fig. 3, GDP changes from the baseline scenario are displayed by the colored bars and those for the test scenario are shown with the transparent bars. For both scenarios, GDP progressively declined as a function of disruption severity. At the same time, we note that changes in GDP due to travel time delays were significantly more consequential than for their baseline counterparts. For example, in this model, a random disruption of 3% of road segments in the San Francisco urban area results in a $c(\rho) = 34\%$ transportation cost increase. This leads to a 6.64% GDP decrease, significantly more than the baseline result of 0.64%. Not all cities show such disparate results between the two scenarios. For example, for Los Angeles (the 5th bar), a 1% roadway disruption increases the travel time by approximately 1% and therefore the GDP effects are the same (the transparent and the colored bars completely overlap).

Even though higher percent additional delay due to road network disruption predictably results in greater economic impact (Fig. 4), the lack of complete linearity signifies that there may be differences between the city economies in question that lead to the different impacts of regional disruption on economic outputs. For 5% disruption, the proportion of GDP loss in Orlando, Tampa, and Jacksonville are lower than the trend while the rate of GDP loss in San Francisco and Detroit are higher than the trend.

A more detailed study of the relationship of total regional GDP and GDP per capita loss to travel disruption may be important for revealing fundamental differences across studied metropolitan areas. The rate of GDP loss associated with each additional 1% network disruption is not constant within or across cities. Some cities lose a similar amount of GDP with each additional increment of disruption while others lose a more variable amount (see Fig. 5; note that the losses are not displayed cumulatively). For example, Orlando has a similar amount of loss each time another 1% of the roads are disrupted (visualized through tight clusters of graphical points). Houston suffers less loss from 2 to 3% disruption than from 1 to 2% but then much more loss from 3 to 4%, which is visualized as more disperse points. This demonstrates that the impact of additional disruptions is difficult to predict and that there may be thresholds and unknown factors in play that affect whether additional road network disruptions will translate to disproportionate economic impacts.

To investigate whether the tolerance of city economies to road network disruptions is related to their size (as reflected by GDP) or wealth (GDP per capita), we look at how much the forecasted GDP or GDP per capita changes over a 1% disruption increment. The results here are for the disruption window of 4–5%. Loss of GDP over that increment is negatively correlated with GDP (Fig. 6A). That is, larger economies, as measured in GDP, tend to suffer more loss in productivity than smaller economies when road network disruption is increased from 4% to 5%. All cities lose an additional 1% ($\pm 0.4\%$; mean = 0.98%, standard deviation = 0.8%) of GDP with the exception of Los Angeles (0.33% loss) (loss as a percent of GDP is displayed as inset graph). In comparing the main and inset graphs, we can see that the largest economy (Los Angeles) is the most tolerant to network disruption but the next two largest economies (San Francisco and Atlanta) are comparably intolerant, as they lose a similar percent of their GDP as some of the smallest economies in the sample (Tampa and Jacksonville).

Modeled economic impact is also negatively correlated with wealth, as measured in GDP per capita (see Fig. 6B), although to a lesser extent than GDP. Cities with higher productivity per capita have a higher rate of loss when percent road network disruption is increased from 4% to 5% than cities that have lower per capita GDPs. The cities that are more tolerant, in that they lose less GDP per

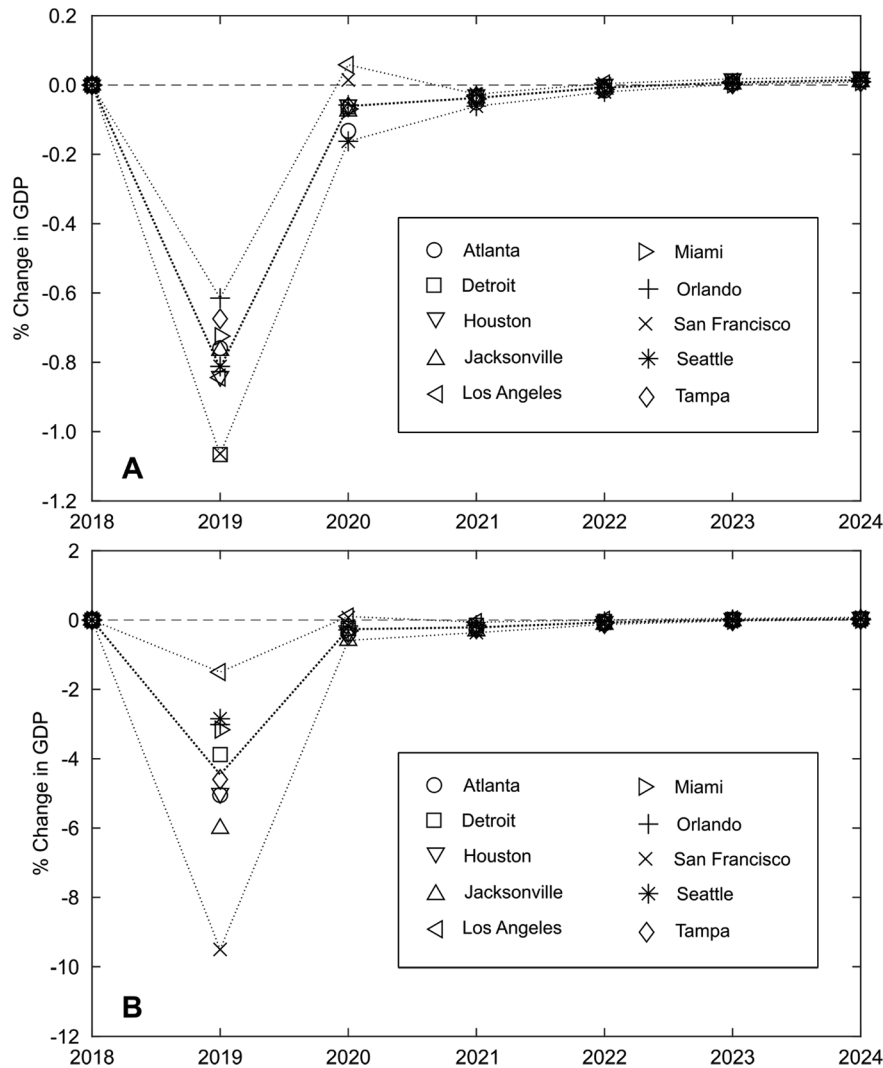


Fig. 2. Temporal performance of regional economies (as measured by GDP) in response to a 5% increase in travel costs that lasted one year (A) and to a 5% transportation network disruption that lasted one year (B). Dotted lines show the lowest, mean, and highest values for each year. The dashed line corresponds to zero change in GDP from expected GDP in the absence of disruption.

capita than their counterparts, have a wide range of wealth; Seattle, Miami, and Orlando lose a similar percent of their GDP per capita, as evident from the inset.

4. Discussion and conclusions

This study is the first to integrate network resilience models for transportation networks with economic models. We found that travel delay resulting from disruptions in transportation networks result in non-trivial economic impacts. The economic outcome of disruptions cannot be predicted by a simple incremental change of velocity-related parameters (VMT, VHT). This research demonstrates that economic models must also be paired with structural network models and analysis in order to best reflect the impact of disruptions on economically important processes, such as the access of commuters to their workplaces and firms moving their commodities to market. This work also shows that the degree of disruption (i.e., percent of roadways disrupted) cannot be assumed to be proportional to transportation cost increase. Rather, network disruptions can result in a wide variety of transportation cost increases depending on a number of different factors. As such, one important conclusion of this work is that losses associated with network disruptions may be an order of magnitude higher than the size of the disruption itself, as evidenced in Fig. 3.

Improving the ability of infrastructure systems to continue to function after unexpected disruptions (i.e., resilience) is emerging as a high priority in infrastructure planning. Resilience is an objective that has to be balanced against other system performance objectives, including the efficiency of transportation networks. Ganin et al. (2017) used a previously developed methodology for network resilience quantification (Ganin et al., 2017) and demonstrated that efficiency and resilience are not correlated in the cases

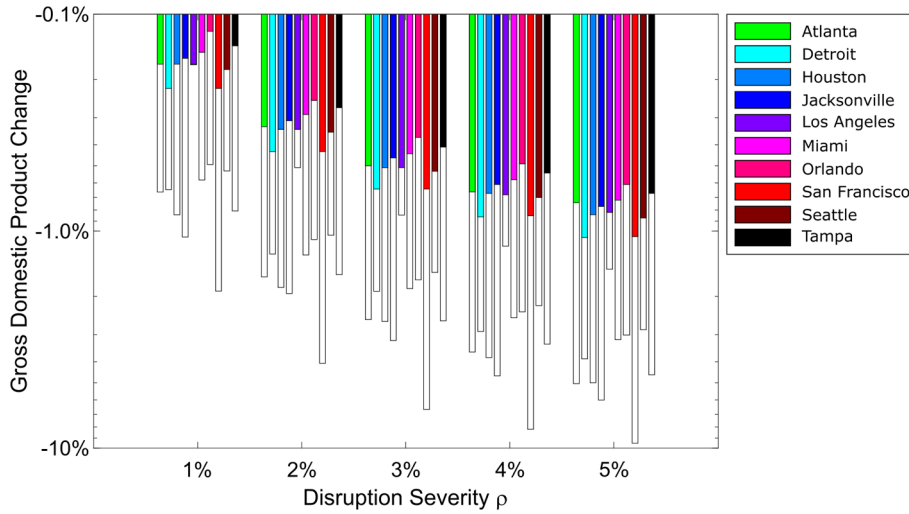


Fig. 3. Impact of disruption on GDP (vertical axis scale is logarithmic). The colored bars correspond to the baseline scenario, with transportation cost increase being proportional to the disruption severity ρ , while the transparent bars present the case when transportation costs change per travel time increase in response to a disruption. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

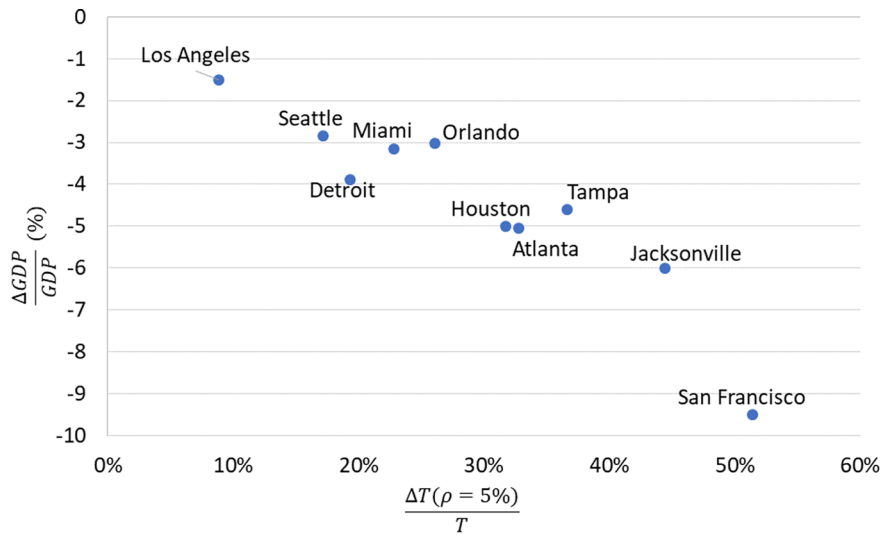


Fig. 4. Percent additional travel time that results from the simulated disruption of 5% of the road network is plotted against the percent loss of GDP that results from that delay. Loss of GDP is related to the delays caused by disruption.

of 40 U.S. cities. Yet, [Ganin et al. \(2017\)](#) did not assess the economic implications of a lack of resilience in transportation networks. [Rose \(2017\)](#) gives a comprehensive overview of the challenges faced in accounting for the economic impacts of disruptions, including that of translating damaged public infrastructure into broader economic losses.

It is intuitive that the effects of disruptive events are not only direct physical impact. The extent to which these impacts can propagate through regional economies needs to be quantified in order to motivate remedial action and inform priorities. A combination of factors is likely contributing to the forecasted impacts of road network disruptions on economies. Potential factors are characteristics of both the road networks as well as the economies. In this proof-of-concept study, only exploratory analysis of selected economic characteristics was used. A more detailed analysis of transportation networks was not attempted.

The question of what makes some cities more sensitive to transportation disruptions than others is important to study further. Our findings show that the economic effects of road disruptions do not perfectly scale with the productivity of city economies (in terms of GDP) or wealth (in terms of GDP per capita) and that not all of the regional economies in the study are equally affected by disruptive events. A naïve expectation is that the more productive an economy, the more sensitive it will be to transportation infrastructure disruption due to its reliance on a mobile workforce. Yet, the results of this study imply that more complex processes are at work. Further evidence of the differing sensitivity of city economies to disruptions include that each additional increment of disruption does

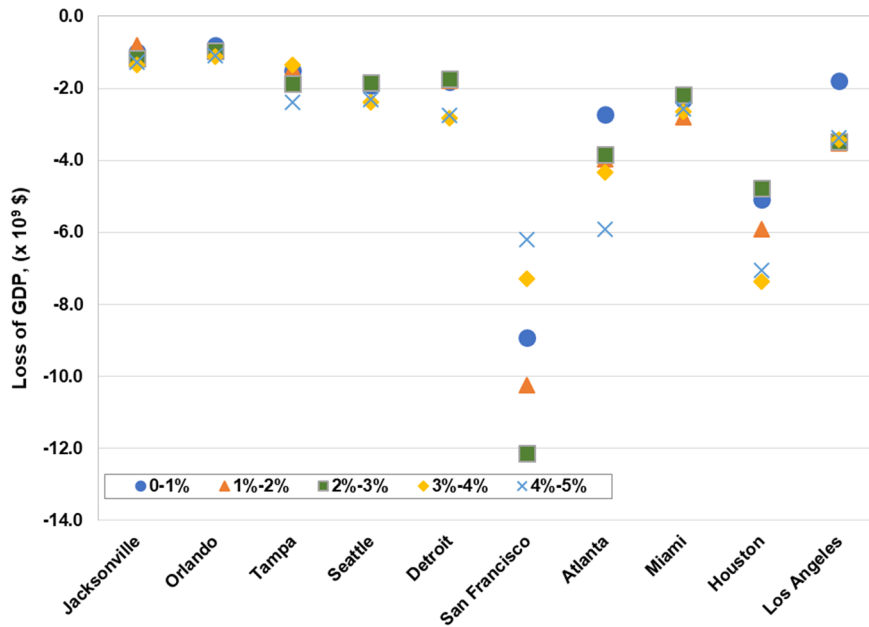


Fig. 5. GDP loss attributable to each additional 1% road network disruption (not cumulative). Losses attributed to an increment of disruption are variable and do not have a pattern that follows the addition of increments.

not show a pattern across cities and that recovery from simulated disruption is not the same across cities. We can speculate that the sensitivity of an economy is dependent on the reaction of the transportation network to a disruption as well as the dependence of the economy on its transportation network, but more research is needed to understand the factors at play. The implication of this result is that both region-specific economic and region-specific transportation models are necessary for resilience planning.

The model is limited, however, in that the calculated transportation cost increase was applied for a period of an entire year; a finer temporal resolution would be preferable and would yield more interesting results in terms of recovery of economies from disruptive events. Additionally, the modeled cost increases may be conservative as, in reality, unexpected delays can result in disproportionately higher cost increases as compared to expected delays. For example, traffic congestion does not disrupt businesses on a day-to-day basis because it can be planned; unforeseen delay cannot be planned around and may cause disruptions that hurt businesses. Because Value of Reliability is not taken into account in the economic model used here, the economic impacts of transportation disruption were conservative. Other limitations include excluding public transportation, which is an important consideration in coupled network models. Similarly, the traffic model is intentionally abstract and simple and only uses readily available data. Integration of explicit travel pattern (e.g. using cell phone data) could improve specificity of the model, but it will limit its universal appeal because these data may not be available for all cities and may be of varying quality. Finally, the current model does not differentiate between the expected travel times, the variance of everyday traffic, and unpredictable delays. Future work should investigate how to integrate potential methods of measuring the Value of Reliability, separate from expected travel times, as well as how unexpected delays may best have their costs modeled and integrated. Similarly, whereas this research considers how different city economies recover from a simple and passive standpoint, future work should consider the economies themselves with a more in-depth consideration of the specific economic indicators and recovery patterns (Rose and Dormady, 2018).

A key motivation of this research is to highlight that efficiency, risk reduction, and resilience are different and often competing objectives as it relates to making investment decisions on improving transportation networks. Although disruptive events result in delay, and sometimes much more pronounced delay than routine congestion, they cannot be prevented solely by improving efficiency of a transportation network or its hardening to reduce risks. Risk reduction can have limited success for uncertain events. For example, in flow networks, such as transportation systems, efficiency may be achieved by having sufficient roadway capacity while resilience can result from the availability of (potentially not very efficient) alternative routes, which would ensure graceful performance degradation under disruption as opposed to a complete collapse. Planning for resilience is functionally different than planning for efficiency and hardening (Ganin et al., 2017).

Future work should explore the driving factors of uneven outcomes across cities, namely the reasons for transportation network sensitivities to disruption as well as local economy sensitivities to transportation failures and ability to recover. Additionally, future work can advance areas in which this study were limited and move the method toward practical application in transportation planning. For example, this paper did not investigate the topological attributes of road networks that yield more or less delay. This paper also did not investigate the mechanisms by which network disruptions cause economic impact in the results. Efforts to enhance the resilience of road networks to disruption and/or of the regional economy to a lack of road network resilience will need to study the outcomes of the models in detail. This paper is the first to provide fusion of a well-established Regional Economic Model (TranSight 4.2 User Guide, 2018) with the transportation network resilience model developed by Ganin et al (2017), and to apply it

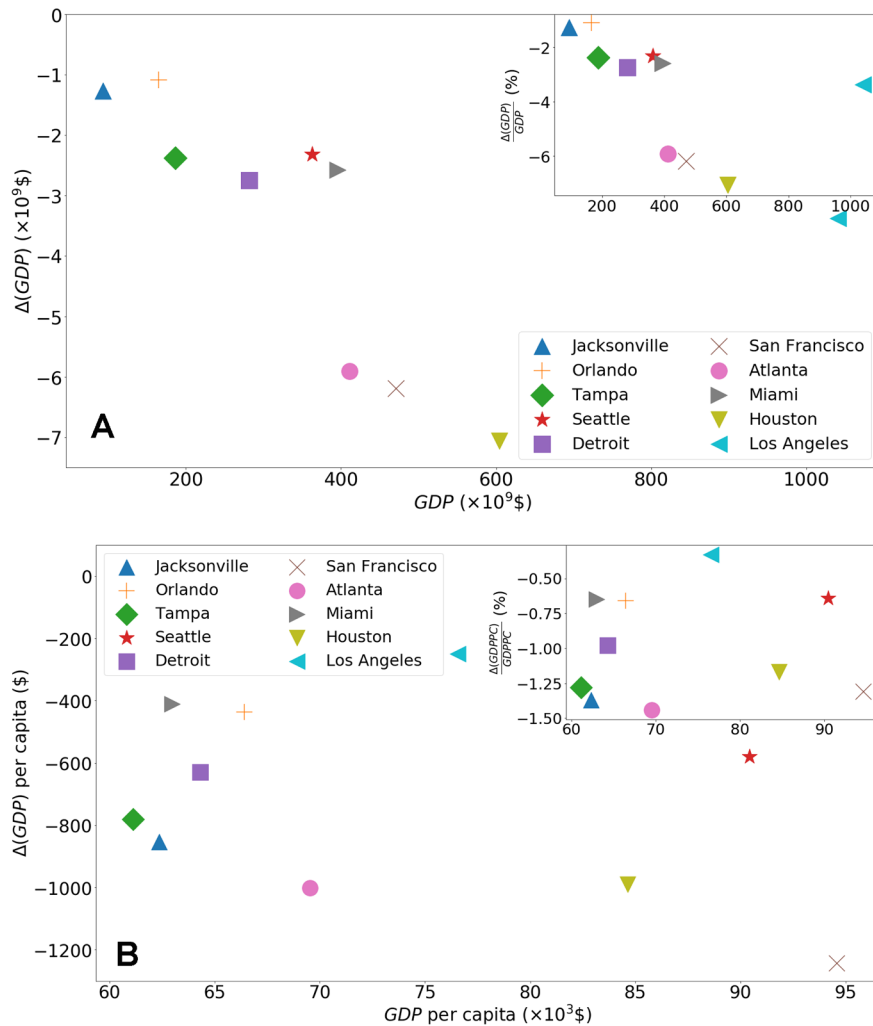


Fig. 6. Comparison of city economy size (as measured by GDP) (A) and wealth (as measured by GDP per capita) (B) as explanatory variables of the tolerance of city economies to additional disruption. Results are for additional GDP losses attributable to increasing percent network disruption from 4% to 5%. Insets show losses as a percent of GDP and GDP per capita, respectively.

to multiple cities in the United States. This explicit integration of a temporal resilience model linked with a regional economic model allows for the comparative evaluation of multiple cities with respect to the impact of resilience. This paper demonstrate that neglecting to plan for disruptive events from a resilience perspective can result in a disproportionate impact on city economies.

Authors statement

Igor Linkov and Maksim Kitsak developed ideas and approaches. Margaret Kurth led paper preparation and interpretation. Alex Ganin was responsible for resilience modeling. William Kozlowski, Billy Leung, and Jeffrey Dykes provided REMI modeling. Avi Mersky contributed to model development.

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